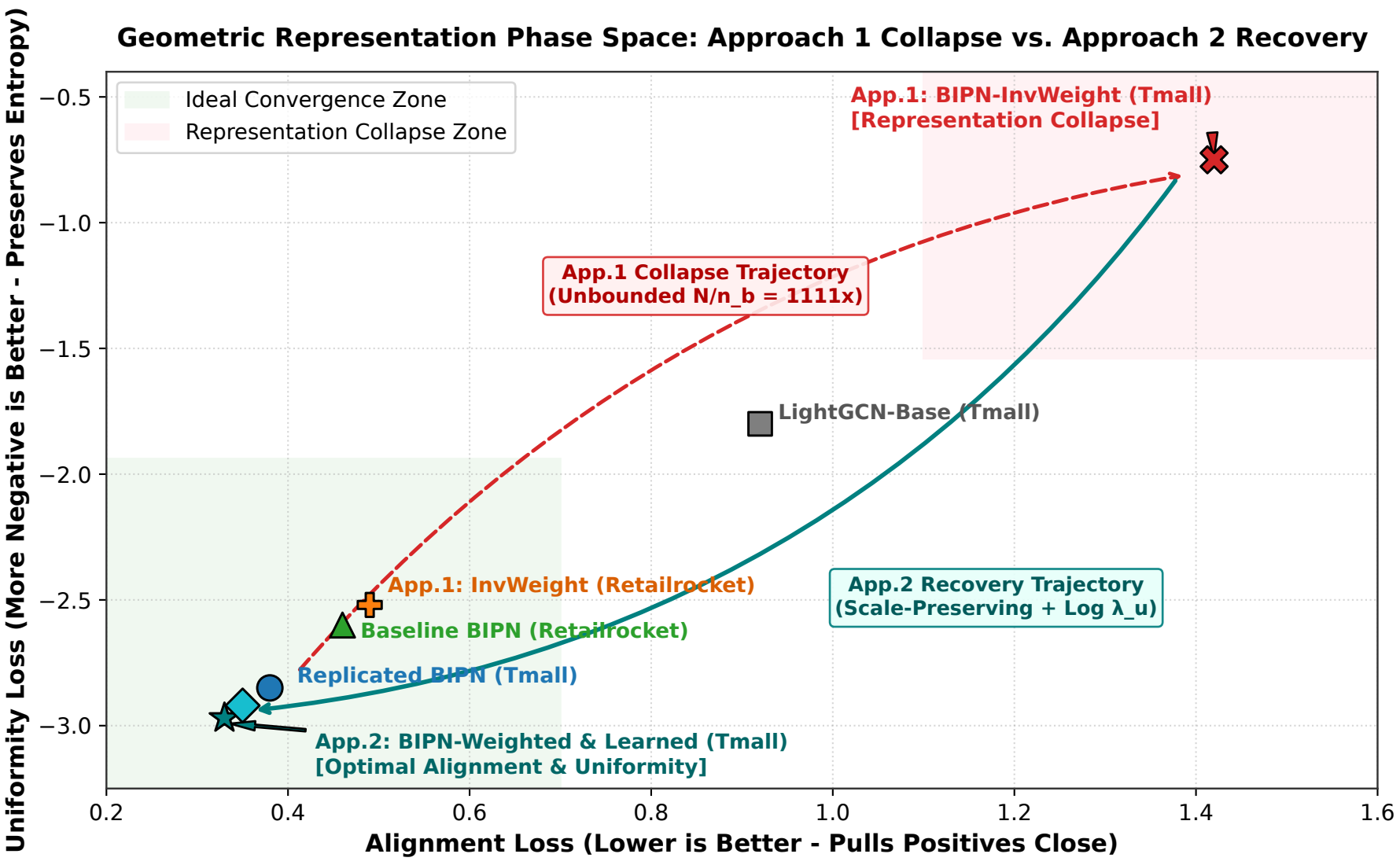
Geometric Representation Phase Space: Approach 1 Collapse vs. Approach 2 Recovery
Ideal Convergence Zone
Representation Collapse Zone
App.1: BIPN-InvWeight (Tmall)
[Representation Collapse]
App.1 Collapse Trajectory
(Unbounded N/n_b = 1111x)
LightGCN-Base (Tmall)
App.1: InvWeight (Retailrocket)
Baseline BIPN (Retailrocket)
Replicated BIPN (Tmall)
App.2 Recovery Trajectory
(Scale-Preserving + Log λ_u)
App.2: BIPN-Weighted & Learned (Tmall)
[Optimal Alignment & Uniformity]
Uniformity Loss (More Negative is Better - Preserves Entropy)
Alignment Loss (Lower is Better - Pulls Positives Close)
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6